

Traversal in a BT :

Traversing a BT means to visit each vertex/node of the BT.

Here we have three types of traversal rule of a BT :

- i) Preorder Traversal (Root - L - R)
- ii) Inorder Traversal (L - Root - R)
- iii) Postorder Traversal (L - R - Root)

Algorithm :

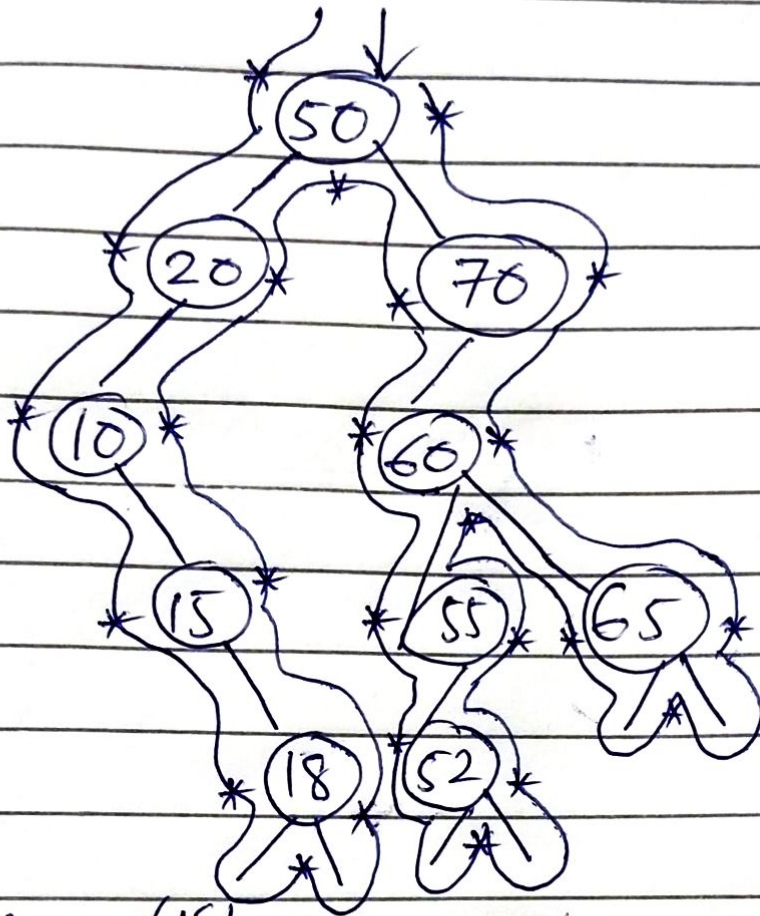
- i) Preorder -
 - a) Root node visit first
 - b) Traverse Left subtree
 - c) Traverse Right subtree
- ii) Inorder -
 - a) Traverse Left subtree
 - b) Visit Root node
 - c) Traverse Right subtree
- iii) Postorder -
 - a) Traverse Left subtree
 - b) Traverse Right subtree
 - c) Visit Root node.

Short cut Method -

- i) Preorder : 1st time visited nodes
- ii) Inorder : 2nd time visited nodes
- iii) Postorder : 3rd time visited nodes.

Example: Find the preorder, inorder and postorder expression/sequence of the BST: 50, 20, 10, 70, 60, 15, 18, 65, 55, 52

Solⁿ:



Preorder : (1st time visited nodes)
50, 20, 10, 15, 18, 70, 60, 55, 52, 65

Inorder : (2nd time visited nodes)
18, 15, 10, 20, 50, 52, 55, 60, 65, 70

Postorder : (3rd time visited nodes)
18, 52, 65, 60, 50